Abdallah AbdelMomen Abdallah Shehawey

Embedded Software Engineer

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Professional Summary

Highly motivated Embedded Software Engineer with extensive experience in developing and optimizing embedded systems. Proficient in C/C++ programming, real-time operating systems (RTOS), and hardware-software integration. Expertise in microcontroller programming, communication protocols (UART, SPI, I2C, CAN, LIN), and debugging. Strong background in AUTOSAR, data structures, and algorithms. Passionate about delivering innovative solutions for the automotive and embedded systems industries.

Education

Student in Electronics and Communication Engineering **Oct 2021 – Jul 2026**
Al-Azhar University **Cairo, Egypt**
Expected Graduation: [Jul-2026]
Core Areas: Embedded Systems, Electrical Power Engineering, Software Development, Control System

Key Skills

- **Programming Languages:** C, C++, Embedded C, Python, Assembly, Verilog.
- **Embedded Systems:** RTOS, Microcontroller Interfacing, Device Drivers (RCC, GPIO, ADC, UART, I2C, SPI, PWM, DMA, NVIC, SYSTICK).
- **Embedded Linux:** Linux Administration, SPL
- **Development Tools:** Atmel Studio, STM32 CubeIDE, VS Code, Eclipse,Proteus, Keil µVision.
- **Debugging & Optimization:** Troubleshooting, Data Structures, Algorithms, Real-Time Debugging Tools, Static Code Analysis.
- **Communication Protocols**: UART, SPI, I2C, CAN Protocol, LIN Protocol, Ethernet.
- **Software Development:** AUTOSAR Fundamentals, Embedded Linux, Unit Testing.
- **Version Control:** Git & GitHub.
- **Additional Tools:** MATLAB, JTAG Debugging, Real-Time Trace Tools, Logic Analyzers, Oscilloscopes.
- **Hardware Proficiency:** ARM Cortex-M, AVR, PIC, ESP32, Raspberry Pi.
- **Soft Skills:** Problem Solving, Team Collaboration, Leadership, Adaptability, Structured Thinking.

Volunteering Experience

- **Vice Head of Embedded System - AZEX** **Oct 2024 – May 2025**
  Al-Azhar University, Cairo, Egypt
  - Supervised a team 0f 10+ engineers, enhancing technical capabilities by 20% through structured training programs.
  - Delivered advanced training on RTOS concepts and microcontroller programming.
  - Designed and implemented training modules covering real-time applications and embedded systems.
- **Core Team of Embedded System – GDSC** **Apr 2024 – Sep 2024**
  Google Developer Student Club, Al-Azhar University, Cairo, Egypt
  - Led hands-on workshops on embedded systems development.
  - Delivered advanced training on RTOS concepts and microcontroller programming.
  - Organized Coding Competitions, fostering Collaboration and practical learning among participants.
- **Instructor & Technical Support - AZEX** **Mar 2024 – Sep 2024**
  Al-Azhar University, Cairo, Egypt
  - Delivered lectures on C programming to university students, increasing comprehension rates by 95%.
  - Coordinated technical workshops, ensuring the successful participation of over 200 students.

Certifications and Training

- **Embedded System Diploma** **Sep 2023 – Mar 2024**
  Eng. Mohamed Zaghloul, Eng. Mahmoud Karem, Eng. Ahmed Abdel Ghafar, NTI
  During this diploma studied these topics:
  - Advanced C programming, Data Structures, and Algorithms.
  - Developed device drivers for ATmega family microcontrollers (DIO, ADC, UART, I2C, SPI, PWM).
  - Practical implementation of RTOS concepts based on AVR microcontrollers.
  - Gained foundational knowledge of the AUTOSAR layered architecture.

- **Problem Solving**
  Hacker Rank: https://www.hackerrank.com/abdallahShehawey
  - Earned 5-star rating in C programming on Hacker Rank.
  - Strong skills in algorithms, data structures, and code optimization.
  - Experienced in debugging and solving complex programming challenges.
- **Advanced Embedded Systems Diploma** **Jul 2024 – Present**
  By Eng. Amgad Samir – Senior Embedded Software Engineer at Valeo. **Hours: 268**
  Completed The Following:
  - STM32 architecture, ARM Cortex-M fundamentals, and development of drivers (RCC, GPIO, UART, SPI, I2C, DMA, ADC, Timers).
  - Gained hands-on experience in debugging, interrupts, low-power modes, Git, and CAN/LIN communication
  - Upcoming: Advanced RTOS, Bootloader development, and AUTOSAR fundamentals.
- **Embedded Systems ARM Track "Summer Training"** **Jul 2025 – Aug 2025**
  Information Technology Institute (ITI), Egypt **Hours: 150**
  During this training studied these topics:
  - ARM Architecture, Peripherals, Interrupts.
  - Real-Time Concepts, Acoustic Signal Generation, Communications, IOT, Bootloader.
  - Graduation Project.
- **Embedded Linux** **Jul 2025 – Present**
  Currently Studying Embedded Linux, including system architecture, build system architecture, build systems, toolchains, and kernel basics.
  During this diploma studied these topics:
  - Linux boot process, system services management (systemd), and user/group administration.
  - File permissions, access control, and filesystem hierarchy.
  - Package management, shell scripting, disks & partitions, and system monitoring tools.
  - Networking configuration, firewalls, and basic security hardening.
- **System Programming under Linux** **Jul 2025 – Present**
  STMicroelectronics **6 Months**
  - Building Linux utilities from scratch (cp, mv, pwd, ls, echo, custom CLI tools).
  - Implementing multiple custom shells (femto, pico, nano, micro shells).
  - Filesystem management and Linux file system operations.
  - Process management and building a minimal Linux distribution and testing it on QEMU.
  - Networking fundamentals, socket programming, and Linux networking CLI.
  - Introduction to Yocto build system and embedded Linux routing.
  - Multithreading and advanced Linux system programming concepts.

Projects

- **Smart Home Project:** Developed a fully integrated home automation system.
- **RTC Watch:** Engineered a real-time clock system with precision timing.
- **MP3 Player:** Built an MP3 player using ARM Cortex-M4 with smooth audio playback**.**
- **Electronic Voting Machine:** Built a secure voting platform with real-time data processing.
- **Advanced Safe System:** Developed a secure access-controlled safe leveraging embedded security algorithm.
- **Traffic Light Controller:** Created a real-time traffic management system, reducing delays by 15%.
- **Debugging Tool:** Designed a tool that reduced testing time by 30%, improving overall development efficiency.
- **Simple Calculator:** Programmed a user-friendly calculator
- **Instructions Decoder:** Developed a system to decode and execute custom instructions efficiently
- **CAN Bus Analyzer:** Implemented a tool to monitor and debug CAN bus communication, enhancing diagnostics efficiency
- **Firmware Optimization:** Reduced memory usage by 25% and improved execution speed by 30% in existing embedded applications
- **IoT Weather Station:** Built an IoT-based weather station using ESP32 and cloud services
- **Battery Management System:** Designed an efficient power monitoring system for lithium-ion batteries

Language

- **Arabic:** Native
- **English:** Very good

Personal Information

- **Date of Birth:** 19 / 5 / 2003
- **Gender:** Male
- **Nationality:** Egyptian